

Community-based Health Insurance and Health: Evidence from Senegal *

Ibou Dieye[†]

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Abstract

Community-based health insurance (CBHI) schemes have been promoted in low- and middle-income countries as a means to extend financial protection and healthcare access to populations excluded from formal insurance. Existing studies suggest modest gains in utilization and reduced out-of-pocket expenditures, but evidence on health outcomes is sparse and most evaluations rely on observational comparisons. This paper provides the first nationwide, quasi-experimental evaluation of CBHI in Senegal, where 676 mutual health organizations (MHOs) were introduced across departments between 1990 and 2019. I link administrative rollout data with eleven Demographic and Health Surveys and five Service Provision Assessment surveys, and exploit staggered implementation using difference-in-differences estimators. I trace effects along the causal chain from insurance coverage to financial protection, service utilization, and health outcomes. CBHI increased maternal insurance coverage and reduced the likelihood of high out-of-pocket expenditures, with modest improvements in delivery care quality concentrated in urban areas. I find no consistent effects on other dimensions of utilization, pregnancy loss, or neonatal mortality. These results highlight both the potential and the limits of voluntary CBHI: while schemes can reduce financial barriers, weak demand for insurance and limited supply-side responsiveness undermine their effectiveness. Achieving universal health coverage will require more effective financing strategies and complementary investments in service quality.

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[†]Harvard University. ibrahimadieye@g.harvard.edu

1 Introduction

Universal Health Coverage (UHC) is widely recognized as a cornerstone of equitable and resilient health systems. First articulated in the Alma-Ata Declaration of 1978, its importance has been reaffirmed through successive global agreements and policy frameworks (Baum, 2007). More recently, UHC has been enshrined in the 2030 Sustainable Development Goals (SDGs), with Target 3.8 committing countries to ensure access to essential health services and financial protection (United Nations, 2015). Achieving these goals requires financing models that reach populations historically excluded from formal insurance, particularly in low- and middle-income countries (LMICs).

Despite broad consensus, progress toward UHC has been uneven, especially in LMICs where health systems are characterized by high out-of-pocket (OOP) payments and limited risk pooling (World Health Organization, 2023). A central barrier is the limited reach of formal health insurance, which often excludes the informal sector, estimated to account for 80–90% of the workforce in many LMICs (International Labour Organization, 2018). In response, governments and development partners have promoted community-based health insurance (CBHI) schemes, known in West Africa as mutual health organizations (MHOs). These voluntary arrangements pool community resources to provide financial protection and improve healthcare access (Chemouni, 2018; Alenda-Demoutiez and Boidin, 2019). CBHI has long been viewed as a promising approach for extending coverage in rural and underserved areas where formal schemes are absent (Dror and Jacquier, 1999), but recent evidence has raised questions about its effectiveness and scalability.

A large body of research has examined the effects of CBHI, with most studies focusing on intermediate outcomes. Many find modest increases in healthcare utilization and reductions in OOP expenditures (Eze et al., 2023). Far fewer studies assess health impacts. Two notable studies in China and India report improvements in health associated with CBHI (Wang et al., 2009; Aggarwal, 2010), but replication outside these settings has been limited. Taken together, two important gaps remain in the literature. First, most evaluations rely on observational comparisons between insured and uninsured individuals. Although many attempt to adjust for selection bias using covariate controls or matching methods (Jütting, 2004; Smith and Sulzbach, 2008), concerns about endogeneity persist, particularly unobserved differences in risk preferences, income shocks, or baseline health that influence both enrollment and outcomes (Degroote et al., 2020). Second, relatively few studies move beyond financial protection and utilization to assess whether CBHI improves health outcomes. Yet improved population health is the ultimate aim of UHC, making this a critical area of inquiry.

Senegal provides a particularly useful setting to address these gaps. Unlike many countries where CBHI remains small-scale or fragmented, Senegal launched a nationwide expansion of MHOs as part of its UHC strategy beginning in 2013 (Daff, 2021). Between 2013 and 2018, the number of CBHI beneficiaries grew from 421,670 to over 3 million (ANACMU, 2019a). During 2015–2018, the government allocated 7.4 billion CFA francs (about USD 12.3 million) in premium subsidies to MHOs, nearly 10% of all subsidies for health utilization in that period (Cour des Comptes, 2019). Despite this substantial investment, no prior study has established a causal link between CBHI coverage and health outcomes in Senegal. Existing evaluations have typically been cross-sectional and geographically limited. For example, Jütting (2004) found reduced hospital expenditures among insured households in Thiès but noted lower enrollment among poorer households. Chankova et al. (2008), using multi-country data, reported limited improvements in care-seeking and modest protection against catastrophic costs. Smith and Sulzbach (2008) documented higher maternal health-care utilization conditional on coverage in underserved areas. More recent evaluations within the national UHC framework found benefits concentrated among non-poor enrollees, with limited effects among the poorest populations (Ly et al., 2022a; Bousmah et al., 2022).

This study provides the first comprehensive, nationwide, quasi-experimental evaluation of Senegal’s CBHI program. I assemble and link data from eleven rounds of the Demographic and Health Surveys (DHS), five Service Provision Assessment (SPA) surveys, and newly compiled administrative records documenting the rollout of 676 MHOs across all Senegalese departments.¹ This integrated dataset allows me to exploit staggered CBHI implementation from 1990 to 2019 and to estimate plausibly causal impacts using state-of-the-art difference-in-differences methods designed for staggered adoption settings. I focus on maternal, reproductive, and neonatal health—domains central to CBHI benefit packages (Rouyard et al., 2022), reliant on sustained contact with skilled providers, and well captured in survey data (Croke et al., 2020). My analysis follows the causal chain from insurance coverage to financial protection, healthcare utilization, and health outcomes, and examines heterogeneity by urban-rural setting to assess whether CBHI reached underserved populations.

My results show that CBHI increased insurance coverage for maternal care by 6.7 percentage points (95% CI: 2.4, 11.1) and reduced the likelihood of top-quintile OOP expenditures by 22.6 percentage points (95% CI: −36.0, −9.1). Exposure improved an index of delivery care quality by 2.8 percentage points (95% CI: 0.04, 5.5), with larger gains in urban areas (4.8 percentage points, 95% CI: 2.0, 7.6). I find no significant effects on other utilization measures, pregnancy loss, or neonatal mortality. While relative gains in coverage are sub-

¹Departments are Senegal’s second-level administrative subdivisions, comparable to districts in other countries.

stantial given low baseline levels, the absolute reach of CBHI remains limited. This pattern is consistent with broader evidence that demand for health insurance is weak even when programs are heavily subsidized (Das and Do, 2023). One explanation, highlighted by Das and Do (2023), is that provider responses can undermine the objectives of insurance schemes by charging above mandated prices, offering unnecessary care, and creating uncertainty about whether benefits will be honored or whether treatment will be adequate. Similar concerns are reflected in national reports of billing fraud among Senegalese providers (Cour des Comptes, 2019).

Against this backdrop, my findings illustrate both the promise and the limits of CBHI as a pathway to UHC in LMICs. Improvements in coverage and financial protection suggest that community-based schemes can reduce cost barriers and increase access to institutional delivery. Yet the modest magnitude of these effects, combined with low enrollment, limited take-up, and weak supply-side accountability, underscores challenges in viability, reach, and sustainability. Moreover, the absence of consistent impacts on broader utilization and health outcomes highlights the difficulty of translating insurance coverage into improved population health without concurrent investments in service availability and quality. The results suggest that voluntary, small-scale CBHI may be insufficient to achieve UHC goals without broader system-level reforms.

This study contributes to two strands of literature: the causal effects of health insurance in developing countries and the more specific evidence on CBHI. Insurance expansions generally increase financial protection and utilization, but evidence on health impacts remains limited (Gruber et al., 2023; Das and Do, 2023; Erlangga et al., 2019). CBHI-specific studies largely echo this pattern (Giedion et al., 2013; Eze et al., 2023). A recurring theme in this literature is the substantial variation in program impacts, reflecting differences in design features such as population targeting, covered services, provider payment arrangements, and administrative procedures. Operationally, CBHI faces persistent challenges, including low enrollment and retention, weak contracting with providers, and high administrative costs (Platteau et al., 2017; Ly et al., 2022b; Rouyard et al., 2022).

I add to this literature with new causal evidence from Senegal, a country that has made substantial investments in expanding public insurance but where rigorous evidence on health impacts remains scarce. By combining multiple nationally representative data sources with administrative rollout records and applying modern econometric methods, I provide new evidence on whether CBHI delivers on its stated objectives. While my primary focus is health outcomes, I begin by examining intermediate outcomes along the causal chain: coverage, financial protection, and service use.

The remainder of the paper is organized as follows. Section 2 provides institutional

background on CBHI in Senegal. Section 3 describes the data. Section 4 outlines my empirical strategy. Section 5 presents results, followed by robustness checks in Section 6. Section 7 concludes.

2 Institutional Setting

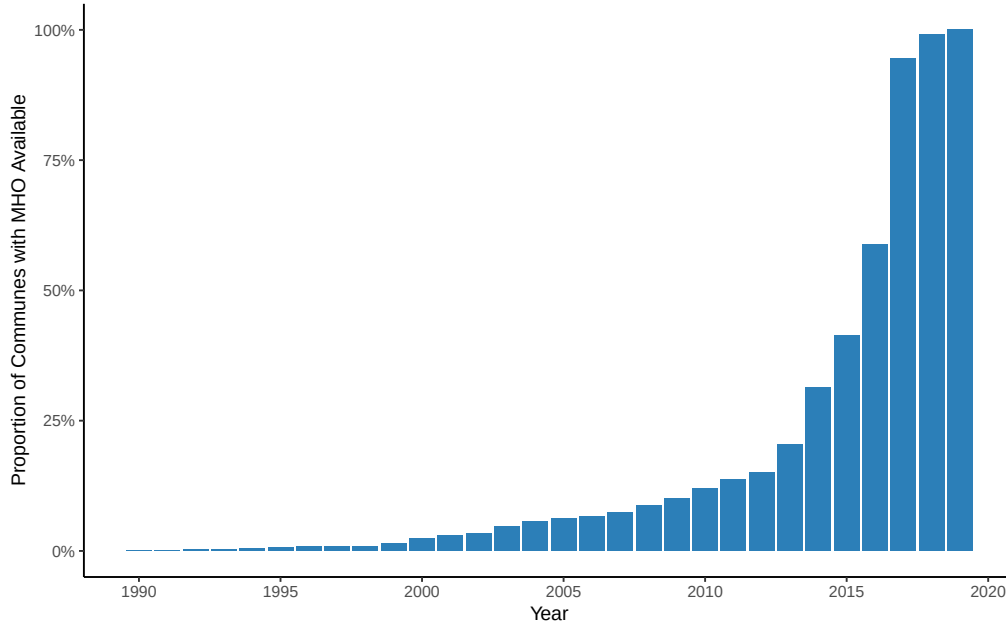
2.1 CBHI Rollout in Senegal

Senegal’s CBHI model began in 1989 with the establishment of the first mutual health organization (MHO) in Fandène (Thiès region), inspired by mutualist practices in France (Jütting, 2004; Alenda-Demoutiez and Boidin, 2019). By 2000, sixteen MHOs were active in the region, covering roughly 30,000 individuals (Jütting, 2004). These early schemes were organized at the community level, with volunteer boards overseeing enrollment, premium collection, and negotiations with local providers.

The Ministry of Health created a dedicated office to promote MHOs in 1998, and by 2008 CBHI had been incorporated into the national strategy for financial risk protection. A major policy shift came in 2013, when CBHI was adopted as a central component of Senegal’s UHC strategy. In 2015, the government established the National Agency for Universal Health Coverage (ANACMU) to standardize and scale implementation nationwide (Daff et al., 2020).

Figure 1 shows the pace of rollout. In 2012, 15% of communes had access to CBHI, increasing to 60% by 2016. By 2019, coverage extended to all communes, making this period central for evaluating Senegal’s CBHI program.

Figure 1. Cumulative Rollout of MHOs Across Communes



Note: This figure shows the cumulative proportion of communes with at least one mutual health organization (MHO) by year. The data are collapsed at the commune level using the earliest year in which an MHO was active.

Under ANACMU oversight, MHOs adopted a uniform benefit package, eligibility criteria, and premium structure. Households paid annual premiums of roughly 7,000 CFA francs (about USD 12) per person, with the government subsidizing 50% for the general population and fully covering premiums for vulnerable groups such as low-income households and persons with disabilities (Daff et al., 2020; Rouyard et al., 2022). The package covered outpatient care, maternal and newborn services, and hospitalization, subsidizing 80% of user fees and generics and 50% of branded pharmaceuticals. Benefits were not portable, however, and enrollees were typically covered only for services at facilities contracted by their local MHO. To expand risk-pooling for high-cost services, MHOs formed unions at the departmental level (Deville et al., 2018; Rouyard et al., 2022).

To address the limitations of small-scale risk pools and volunteer management, the government piloted Departmental Health Insurance Units (UDAMs) in Kounghoul and Foundiougne between 2014 and 2017 (Daff et al., 2020). These professionally managed schemes operated at the department level, aiming to improve efficiency and expand coverage. Encouraged by early results, Senegal began scaling UDAMs in 2020 (Ridde et al., 2022). Because data on UDAMs remain limited and my focus is on the pre-2020 period, I exclude the two pilot departments and analyze only the commune-level CBHI model implemented across the remaining 43 departments.

2.2 CBHI Access and Health

CBHI may influence health outcomes through several channels. By reducing out-of-pocket costs, it can lower financial barriers to care and encourage timely use of antenatal, delivery, and postnatal services (Smith and Sulzbach, 2008). Enrollment may also foster engagement with the health system, supporting preventive care and continuity of treatment. On the supply-side, CBHI can affect provider behavior through contractual arrangements, which may lead to improved care quality. In practice, however, evidence from low- and middle-income settings points to different dynamics. Das and Do (2023) shows that providers may respond to insurance by raising prices, delivering unnecessary care, or creating uncertainty about whether benefits will be honored, thereby undermining the objectives of insurance schemes.

The presence of CBHI schemes in a community may also generate spillovers. Outreach and sensitization efforts can increase community awareness of health risks, the value of insurance, and the importance of timely care-seeking even among non-enrolled individuals. Such exposure may encourage uptake of alternative coverage options (e.g., profession-based schemes) or improved health-seeking behavior more broadly.

Using newly compiled administrative data on the timing and geographic rollout of MHOs, I exploit staggered implementation across departments to estimate causal effects on insurance coverage, financial protection, healthcare utilization, and maternal and neonatal health outcomes. By focusing on community-managed MHOs and excluding departmental UDAMs, my analysis isolates the impacts of Senegal’s flagship CBHI model during its nationwide expansion.

3 Data

3.1 Data Sources

This study combines four primary data sources: administrative records from ANACMU, nationally representative household data from the DHS, healthcare facility data from the SPA surveys, and population estimates from Senegal’s National Agency of Statistics and Demography (ANSD). Table 1 summarizes the years covered and the key measures drawn from each source. DHS surveys, which span the full study period (1992–2019), provide information on insurance coverage, healthcare utilization, and health outcomes. SPA surveys, available for 2012–2017, complement these with facility-level data on insurance and out-of-pocket costs, though the shorter window restricts cost-related analyses.

3.1.1 ANACMU

I use administrative records from ANACMU to determine the timing of CBHI rollout. The dataset documents 676 MHOs across 552 communes as of 2019, of which 666 are community-based MHOs and 10 are professional MHOs targeting occupational groups. I exclude the latter from my analysis.

For most MHOs (601 of 666, or 91%), the dataset records the year of operational launch, which I use as the relevant exposure date because it best reflects when coverage could begin affecting outcomes. The average lag between creation and launch is 0.8 years. For 49 MHOs lacking launch years, I substitute the creation year. For the 16 MHOs with no date information (8 in Tivaouane, 2 in Thiès, 3 in Mbour, and 3 in Fatick), I impute rollout dates using the commune average when available, or otherwise the department-level average. I conduct robustness checks excluding departments with any imputed launch years. In addition, I drop 26 MHOs located in the departments of Foundiougne and Kounghoul, which piloted departmental CBHI units (UDAMs) beginning in 2014.

3.1.2 DHS

I use eleven DHS rounds conducted in Senegal between 1992 and 2019. These nationally representative surveys provide rich data on health, healthcare utilization, and demographic characteristics (Corsi et al., 2012; Demographic and Health Surveys Program, 2022). I focus on three modules:

Birth history module. Retrospective data on all children ever born to women aged 15–49, including survival status and maternal care. The unit of observation is the birth.

Women’s interview module. Individual-level data on family planning, pregnancy, insurance, healthcare utilization, and socioeconomic variables. The unit of observation is the woman.

Contraceptive calendar. Monthly histories of reproductive events and contraceptive use over the 5–7 years preceding the interview, allowing reconstruction of pregnancy episodes and pregnancy loss.

All DHS rounds include georeferenced cluster coordinates, which I use to assign CBHI exposure based on departmental rollout (see Section 3.2). These surveys provide my primary outcome measures on maternal, reproductive, and neonatal health.

3.1.3 SPA

Five SPA surveys conducted between 2012 and 2017 offer nationally representative or near-census assessments of healthcare facility capacity (Kruk et al., 2017; Demographic and Health

[Surveys Program, 2022](#)). Data collection instruments include facility audits, provider interviews, direct clinical observations, and exit interviews. I rely primarily on exit interviews with women seeking antenatal care (ANC) and family planning (FP) services, which provide individual-level data on insurance coverage and out-of-pocket costs. These data complement the DHS by offering direct measures of financial protection during the 2012–2017 period.

3.1.4 ANSD Population Estimates

Commune-level population estimates are drawn from the 2002 and 2013 censuses and official projections for 2014–2019. Because of administrative redistricting, the number of communes increased over time, and some communes listed after 2013 did not exist in 2002.

To generate consistent annual commune-level population estimates from 1990 to 2019, I proceed in two steps. First, I align commune identifiers across years, merging available counts from the 2002 census and subsequent projections. Second, I interpolate gaps, especially for years prior to 2002 and between 2003 and 2012, using natural splines to accommodate flexible growth patterns ([Siegel and Swanson, 2004](#)). Validation exercises show minimal prediction error relative to observed census counts (Appendix Figure [A1](#)). These estimates are used in constructing CBHI exposure variables (see Section [3.2](#)).

Table 1. Summary of Data Sources

Data Source	Years	Survey Modules	Measures Extracted
ANACMU	1990–2019	–	MHO initiation and launch year
DHS	1992–2019	• Women’s interview • Birth history • Contraceptive calendar • Geocoordinates	• Insurance coverage • FP uptake • Antenatal care • Delivery care • Pregnancy outcomes • Neonatal health
SPA	2012–2017	• Exit interviews (ANC and FP) • Facility audits and interviews • Geocoordinates	• Insurance coverage • OOP costs
ANSD	2002, 2013–2019	–	Commune-level population data

Abbreviations: DHS = Demographic and Health Surveys; SPA = Service Provision Assessment; ANSD = National Agency of Statistics and Demography of Senegal; ANACMU = National Agency for Universal Health Coverage; MHO = Mutual Health Organization; ANC = Antenatal Care; FP = Family Planning; OOP = Out-of-pocket.

Note: This table summarizes the data sources used in the analysis, including survey years, modules, and key measures extracted.

3.2 Exposure Measures

I construct CBHI exposure measures by combining administrative rollout data on MHOs with GPS-located DHS and SPA surveys. Because DHS cluster coordinates are randomly displaced (up to 2 km in urban areas, 5 km in rural areas, and up to 10 km for 1% of rural clusters) ([Demographic and Health Surveys Program, 2024](#)), commune-level assignment is unreliable. I therefore assign CBHI exposure at the department level (administrative level 2), where displacement is constrained and locations are more accurately captured. In contrast, SPA coordinates are not displaced, allowing precise commune-level matching in facility analyses.

DHS clusters are mapped to departments using contemporaneous shapefiles from Senegal’s National Agency for Land Use Planning (ANAT) ([United Nations, 2024](#)) and the `sf` package in R ([Pebesma, 2018](#)). Since 2009, DHS displacement has been restricted within

departmental boundaries. As a robustness check, I also re-estimate results restricting the analysis to post-2009 rounds only.

Within each department, MHOs launched in staggered years. To approximate the timing when the typical resident first gained access, I compute a population-weighted average launch year, using commune population in the year prior to each launch. This ensures weights are not influenced by rollout itself. Although reference years differ across communes, the measure reflects contemporaneous exposure. Results are robust to alternative specifications using fixed reference-year populations.

For illustration, Table 2 shows data from Gossas, where six MHOs launched between 2014 and 2017. Using pre-launch populations as weights, the department’s exposure year is calculated as 2016.

Table 2. MHO Launch Years and Pre-Launch Population in Gossas

Department	Commune	MHO Launch Year	Population (Year-1)
Gossas	Gossas	2014	12,698
Gossas	Ndiene Lagane	2016	12,652
Gossas	Colobane	2017	20,172
Gossas	Mbar	2017	30,077
Gossas	Ouadior	2017	11,916
Gossas	Patar Lia	2017	16,221

Formally, letting y_i denote the MHO launch year in commune i and w_i its pre-launch population, the weighted average launch year for department d is:

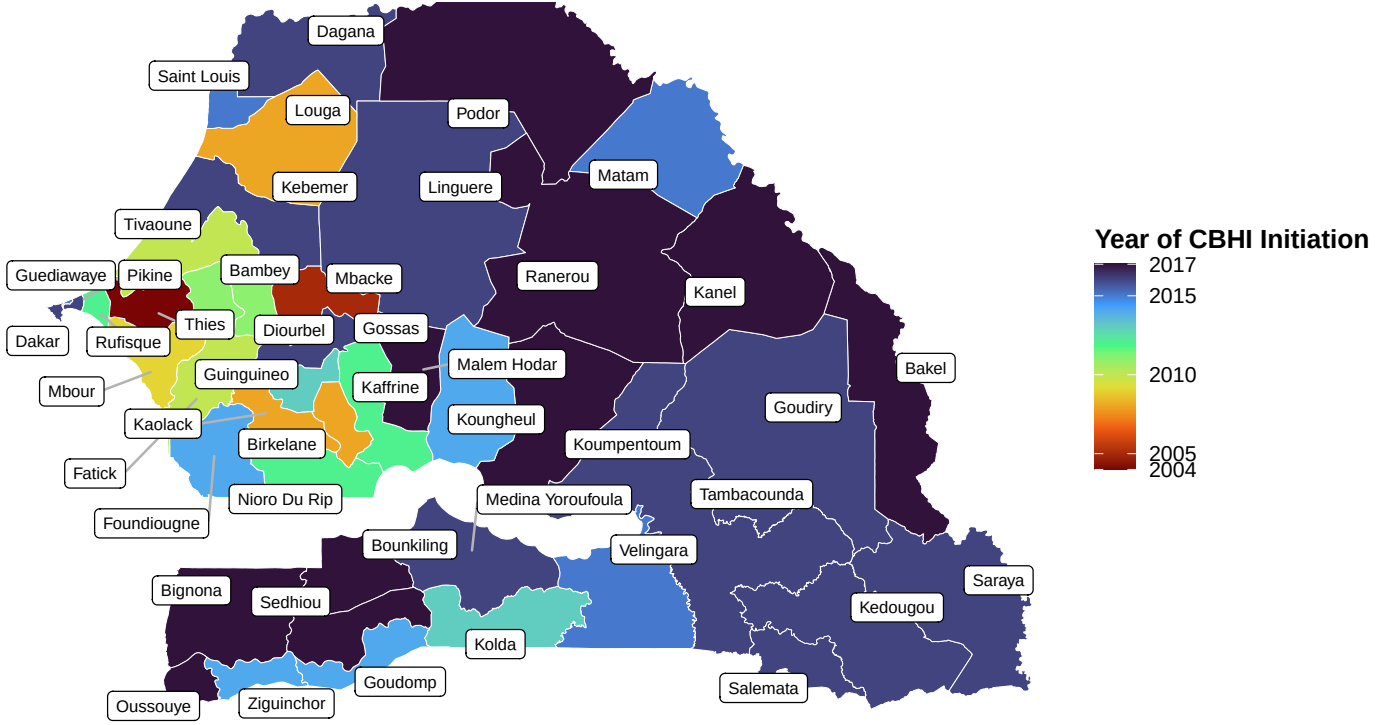
$$\text{Exposure}_d = \left\lfloor \frac{\sum_{i \in d} y_i \cdot w_i}{\sum_{i \in d} w_i} \right\rfloor,$$

where the floor operator assigns the latest full year prior to when the typical resident gained access to CBHI.

This measure improves on a binary indicator based on the first MHO but has limitations. It does not capture within-department intensity or geographic spread, may introduce measurement error that attenuates estimates, and could contaminate pre-treatment periods in difference-in-differences designs due to gradual rollout. I address these concerns in robustness checks and alternative specifications (Sections 4 and 6).

Figure 2 shows the variation in CBHI rollout timing across Senegal’s 45 departments as of 2019, based on my population-weighted definition of departmental exposure. Thiès was the earliest adopter, with exposure beginning in 2004, while ten departments, including Bakel, Podor, and Kanel, did not roll out CBHI until 2017.

Figure 2. Geographic Variation in CBHI Rollout Timing



Note: This figure displays the timing of CBHI exposure across Senegal’s 45 departments (as of 2019). Exposure timing is defined as the population-weighted average MHO launch year across communes within each department.

3.3 Outcome Measures

I group outcomes into four categories that reflect the causal chain from insurance coverage to health: first-stage (insurance uptake), healthcare costs, healthcare utilization, and health outcomes.

3.3.1 First-stage and Healthcare Cost Outcomes

The first-stage outcome is a binary indicator for whether a woman reported having insurance that covered the cost of ANC or FP services at the time of her facility visit. Healthcare costs are measured by a binary indicator equal to one if a woman’s OOP expenditure for ANC or FP services falls within the top quintile of the survey-year cost distribution. This relative measure accounts for inflation and secular changes in user fees and provides a proxy for high OOP costs in the absence of detailed data on catastrophic expenditures. Both outcomes are derived from SPA exit interviews.

Because these outcomes are only observed among women who sought care at a health

facility, estimates capture effects among care users rather than the general population and may be upwardly biased if insurance also increases care-seeking.

3.3.2 Healthcare Utilization Outcomes

I examine both composite indices and individual utilization indicators. To improve statistical power and reduce multiple testing, I construct three indices of care quality and intensity following [Anderson \(2008\)](#):

ANC visit index. Share of three recommended components received: (i) at least four ANC visits, (ii) initiation of ANC by the end of the first trimester, and (iii) care from a skilled provider (doctor, nurse, midwife, or auxiliary midwife).

ANC process index. Share of six recommended ANC services received: blood pressure measurement, blood sample collection, urine sample collection, receipt of iron supplements, at least two doses of tetanus toxoid vaccine, and at least three doses of intermittent preventive treatment for malaria during pregnancy.

Delivery care index. Share of two recommended delivery components: facility delivery and skilled attendance.

Each index ranges from 0 to 1. The selected components align with WHO guidelines for ensuring a positive pregnancy experience ([World Health Organization, 2016](#)).

3.3.3 Health Outcomes

I focus on two health outcomes measured in the DHS.

Pregnancy loss. Defined as any pregnancy ending in a non-live birth, including miscarriage, stillbirth, or abortion, as recorded in the contraceptive calendar.

Neonatal death. Defined as the death of a live-born infant within the first 28 days, using the DHS birth history module.

I restrict attention to neonatal outcomes because children under five were eligible for overlapping free care programs that could confound CBHI effects.

3.4 Additional Measures

I also draw on demographic and socioeconomic characteristics from the DHS and SPA surveys, including age, education, household wealth quintiles, and marital status. These measures describe the study population, serve as covariates in robustness checks, and provide placebo outcomes. All variables are harmonized across survey rounds to ensure comparability over time.

3.5 Summary Statistics

Tables 3, 4, and 5 present summary statistics for demographic and socioeconomic characteristics drawn from the DHS women’s interviews, the DHS birth history module, and SPA exit interviews with women seeking ANC and FP care, respectively. Statistics are shown both overall and stratified by pre- and post-CBHI exposure periods.

In both the DHS and SPA samples, the average age of women is about 28 years. Most women are married, have limited formal education, are not formally employed, and belong to the bottom two household wealth quintiles (Tables 3 and 5). In the DHS birth history sample, the average age at first birth is around 19 years (Table 4).

Across all samples, women surveyed before and after CBHI rollout are broadly similar. Notable differences emerge, however, in education levels and wealth quintiles, which are higher in the post-rollout period. These differences likely reflect broader department-specific socioeconomic changes over time, which I account for in my difference-in-differences framework.

Table 3. Summary Statistics of the DHS Women’s Sample, by Survey Timing

Characteristic	Overall N = 113,542	Survey before CBHI N = 39,891	Survey after CBHI N = 73,651
Age	27.90 (9.34)	27.88 (9.35)	27.91 (9.33)
Education level: no education	58%	63%	55%
Education level: primary	21%	21%	21%
Education level: secondary	20%	15%	22%
Education level: higher than secondary	1.5%	1.0%	1.8%
Married	68%	69%	68%
Employed	44%	43%	44%
Mother age at 1st birth	19.57 (4.03)	19.03 (3.81)	19.88 (4.11)
Education level, husband: no education	76%	78%	75%
Education level, husband: primary	11%	10%	11%
Education level, husband: secondary	9.3%	8.8%	9.7%
Education level, husband: higher than secondary	3.4%	2.9%	3.7%
Age, husband	43.49 (13.92)	43.13 (13.48)	43.62 (14.07)
Residence type	40%	39%	40%
Number of household members	13.67 (8.39)	13.40 (8.41)	13.81 (8.38)
Household wealth quintile: poorest	23%	25%	22%
Household wealth quintile: poorer	23%	25%	22%
Household wealth quintile: middle	23%	24%	23%
Household wealth quintile: middle	18%	15%	19%
Household wealth quintile: middle	14%	11%	15%

Abbreviations: CBHI = community-based health insurance.

Note: Characteristics of women interviewed in the DHS, overall and by timing of interview relative to CBHI exposure. Means and standard deviations (SD) are reported for continuous variables; percentages for categorical variables.

Table 4. Summary Statistics of the DHS Birth Sample, by Pregnancy Timing

Characteristic	Overall N = 306,039	Pregnancy before CBHI N = 228,882	Pregnancy after CBHI N = 77,157
Sex of baby: female	49%	49%	49%
Sex of baby: male	51%	51%	51%
Birth order	3.35 (2.24)	3.28 (2.22)	3.53 (2.29)
Twin birth	3.1%	2.8%	3.9%
Age, mother	35.30 (7.84)	36.23 (7.73)	32.56 (7.50)
Education level, mother: no education	78%	79%	73%
Education level, mother: primary	16%	15%	19%
Education level, mother: secondary	5.8%	5.0%	7.9%
Education level, mother: higher than secondary	0.5%	0.3%	0.8%
Education level, father: no education	81%	82%	79%
Education level, father: primary	9.7%	9.4%	11%
Education level, father: secondary	7.0%	6.7%	7.8%
Education level, father: higher than secondary	2.2%	2.0%	2.8%
Age, father	48.30 (13.00)	49.66 (12.82)	45.15 (12.86)
Residence type	31%	32%	30%
Number of household members	13.90 (8.40)	13.75 (8.51)	14.34 (8.03)
Household wealth quintile: poorest	30%	33%	23%
Household wealth quintile: poorer	26%	27%	25%
Household wealth quintile: middle	21%	21%	23%
Household wealth quintile: middle	14%	13%	17%
Household wealth quintile: middle	8.6%	7.7%	11%

Abbreviations: CBHI = community-based health insurance.

Note: Characteristics of women in the DHS birth history sample, pooling across all births, overall and by timing relative to CBHI exposure. Means and standard deviations (SD) are reported for continuous variables; frequencies and percentages for categorical variables.

Table 5. Summary Statistics of the SPA Sample, by Survey Timing

Characteristic	Overall N = 4,462	Survey before CBHI N = 1,356	Survey after CBHI N = 3,106
Age	27.83 (7.16)	28.08 (7.28)	27.72 (7.11)
Education level: no education	55%	55%	54%
Education level: primary	26%	27%	26%
Education level: secondary	15%	14%	16%
Education level: higher than secondary	4.0%	3.9%	4.1%
Location type: urban	47%	43%	48%
Facility type: health post	66%	66%	66%
Facility type: health center	25%	27%	24%
Facility type: hospital	8.4%	6.9%	9.0%
Facility ownership: public	94%	95%	93%
Facility ownership: non-profit	3.2%	1.8%	3.8%
Facility ownership: private	2.3%	2.1%	2.3%
Facility ownership: faith-based	1.0%	0.7%	1.1%
Service type: antenatal care	45%	44%	46%
Service type: family planning	55%	56%	54%

Abbreviations: CBHI = community-based health insurance; NGO = non-governmental organization; ANC = antenatal care.

Note: Characteristics of women interviewed in the SPA exit survey, overall and by timing relative to CBHI exposure. Means and standard deviations (SD) are reported for continuous variables; frequencies and percentages for categorical variables.

Appendix Table B1 presents SPA statistics stratified by insurance status. Insured women tend to be older, more educated, more urban, and more likely to seek care at private facilities. These patterns are consistent with prior studies showing that individuals who self-enroll in CBHI are typically of higher socioeconomic status than non-enrollees (Jütting, 2004; Conde et al., 2022). In the context of maternal care, such women may also exhibit greater health-seeking behaviors, providing evidence of adverse selection. This underscores the potential for simple comparisons between insured and uninsured groups to overstate the benefits of insurance, even after adjusting for observables. My difference-in-differences design is intended to mitigate these biases.

4 Empirical Strategy

4.1 Baseline Specification

I estimate the impact of CBHI using a difference-in-differences (DiD) framework that exploits variation in the timing of MHO rollout across departments. My baseline two-way fixed effects (TWFE) specification is:

$$Y_{ids} = \beta_0 + \beta_1 \cdot MHO_d + \beta_2 \cdot (\text{post}_{dt} \times MHO_d) + X_i + \delta_d + \phi_t + \gamma_s + \epsilon_{ids}, \quad (1)$$

where Y_{ids} is the outcome for individual i in department d , year t , and survey round s . MHO_d equals one if department d ever receives treatment, and post_{dt} equals one if the observation occurs in or after the first year of CBHI exposure. The vector X_i includes individual-level covariates. Department fixed effects (δ_d) absorb time-invariant differences across departments, year fixed effects (ϕ_t) capture national trends, and survey-round fixed effects (γ_s) account for survey-specific shocks.² Standard errors are clustered at the department level.

This specification compares outcomes within departments before and after CBHI rollout, using not-yet-treated departments as controls. The identifying assumption is that, absent treatment, treated and not-yet-treated departments would have followed parallel trends. While fixed effects address common national shocks and persistent cross-department differences, they do not rule out unobserved shocks that vary over time and across space (for example, localized health initiatives or economic changes). Another concern is reverse causality if rollout decisions were responsive to recent or anticipated outcome changes. My strategy relies on the assumption that rollout timing was not systematically correlated with such shocks,

²In practice, survey year fixed effects are dropped in estimation due to collinearity with year fixed effects.

consistent with historical accounts that emphasize the role of social capital, logistical capacity, donor support, and political factors, rather than local health trends (Alenda-Demoutiez and Boidin, 2019; Wood, 2023). I assess this assumption using event-study tests and examine robustness to alternative specifications in Section 6.

For DHS and SPA outcomes, the time variable corresponds to the survey year. For DHS birth history and contraceptive calendar modules, the time variable refers to the year of pregnancy onset. In the contraceptive calendar, pregnancy start month and year are observed directly. In the birth history module, pregnancy onset is approximated as nine months before the reported birth year. In birth-level analyses, I assume women resided in the same department at childbirth as at survey, consistent with prior work (Croke et al., 2020).

4.2 Estimation Strategy

Because TWFE models may yield biased estimates in settings with staggered adoption and heterogeneous treatment effects (Goodman-Bacon, 2021), my main analysis employs the doubly robust DiD estimator proposed by Callaway and Sant’Anna (2021). This approach avoids negative weighting and allows for treatment effect heterogeneity across cohorts and over time.

I estimate average treatment effects (ATEs) overall and separately by urban and rural strata, using appropriate not-yet-treated comparison groups within each stratum. Dynamic effects are examined with an event-study specification:

$$Y_{idts} = \alpha + \sum_{k=-5}^5 \beta_k \cdot \mathbf{1}\{t - T_d = k\} + X_i + \delta_d + \phi_t + \gamma_s + \epsilon_{idts}, \quad (2)$$

where T_d is the year of CBHI rollout in department d , and $\mathbf{1}\{t - T_d = k\}$ indicates k years relative to rollout, with $k = -1$ omitted. Department, year, and survey-round fixed effects are included, and standard errors are clustered at the department level. To improve stability, leads and lags beyond five years are pooled into $k = -5$ and $k = 5$ bins.

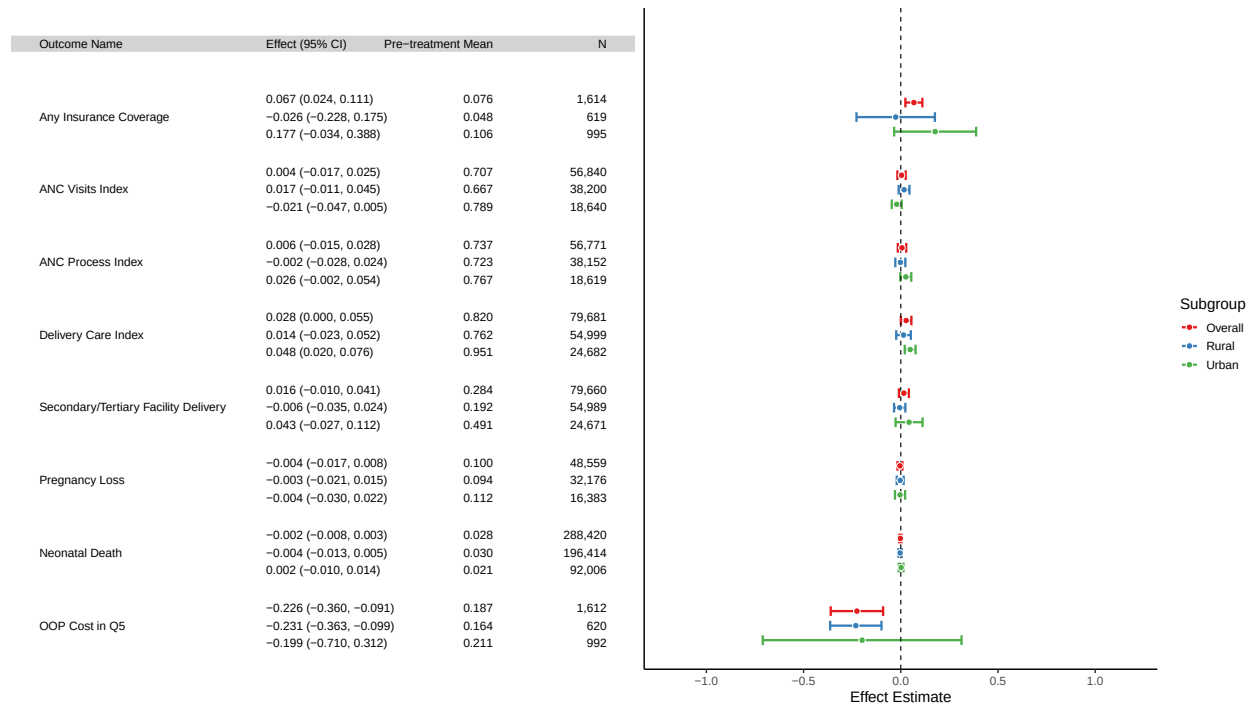
This framework provides a robust and flexible approach to estimating CBHI’s impact, while mitigating the limitations of TWFE in staggered settings. Following best practices, my main specification assumes unconditional parallel trends (Baker et al., 2025) and excludes individual-level covariates, though I test alternative specifications in Section 6.

5 Results

5.1 Main Results

Figure 3 presents ATE estimates from my baseline specification for the overall sample, as well as for rural and urban subpopulations. The corresponding event study plots are shown in Figures 4a–6b.

Figure 3. ATE Estimates



Note: This figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year.

5.1.1 First Stage: Insurance Coverage

Exposure to CBHI increases insurance coverage for ANC or family planning services by 6.7 percentage points (95% CI: 2.4, 11.1) in the overall sample. This represents an 88% increase relative to the low baseline coverage rate of 7.6%. The magnitude of the effect is consistent with administrative data showing that national insurance coverage nearly doubled from

about 20% to 50% between 2013 and 2019 ([ANACMU, 2019b](#)). Baseline rates observed in the SPA surveys align with prior estimates for informal sector populations in Senegal, typically 5–8% ([Schwettmann, 2022](#)).

Subgroup analyses suggest larger effects in urban areas (17.7 percentage points, 95% CI: –3.4, 38.8) than in rural areas (–2.6 percentage points, 95% CI: –22.8, 17.5), though estimates are imprecise given limited sample size. The event study ([Figure 4a](#)) shows no evidence of differential pre-trends, though statistical power is limited.

5.1.2 Healthcare Costs

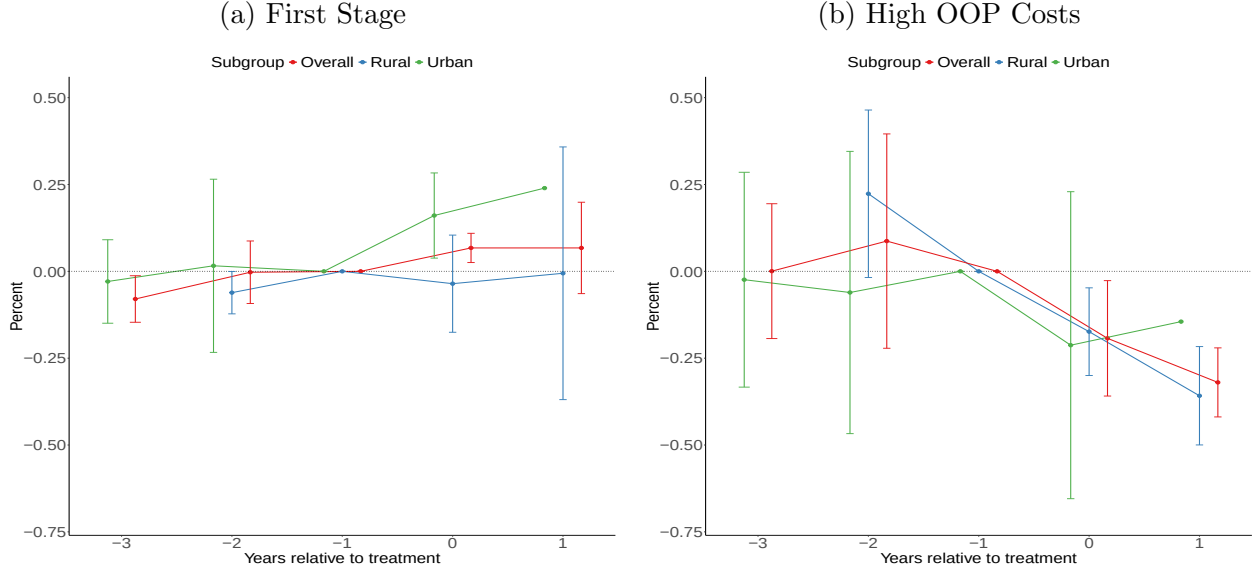
CBHI exposure is associated with a 22.6 percentage point reduction (95% CI: –36.0, –9.1) in the likelihood of incurring high OOP costs. The effect is somewhat larger and more precisely estimated in rural areas (–23.1 percentage points, 95% CI: –36.3, –9.9) than in urban areas (–19.9 percentage points, 95% CI: –71.0, 31.2). Event study results ([Figure 4b](#)) suggest possible divergence in pre-treatment trends in the rural subsample; I return to this in [Section 6](#).

These findings align with earlier evidence from Senegal. [Jütting \(2004\)](#), studying MHOs in Thiès, found that members paid less than half the hospital costs of non-members. [Chankova et al. \(2008\)](#) documented that MHO membership protected against catastrophic hospitalization costs in Ghana, Mali, and Senegal, but not for outpatient care. My SPA sample does not distinguish inpatient from outpatient visits, though 8.4% of respondents reported visiting a hospital and 25% a health center. The maximum OOP payment reported was 40,000 CFA francs (about \$70), suggesting some care may have involved hospital services.

More recent studies highlight equity concerns. [Ly et al. \(2022a\)](#) found that CBHI reduced catastrophic expenditures among non-poor members in three regions (Diourbel, Thiès, Tambacounda), but not among the poorest. Similarly, [Bousmah et al. \(2022\)](#) reported weaker financial protection for fully subsidized CBHI members. Consistent with this, insured women in my SPA sample are older, wealthier, and more urban ([Table B1](#)), indicating the insured subsample is skewed toward the non-poor.

Overall, CBHI reduces the financial burden of high OOP payments among care users. Because SPA data only capture women who sought care, some of whom may have done so because they were insured, estimates may overstate population-wide effects. Nevertheless, these results provide new causal evidence on CBHI’s financial protection role.

Figure 4. Event Study Estimates for the First Stage and Cost Outcomes



Note: This figure presents event-study estimates using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. Both outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. To facilitate comparison across outcomes, the y-axis is standardized to range from -0.75 to 0.5 in each panel.

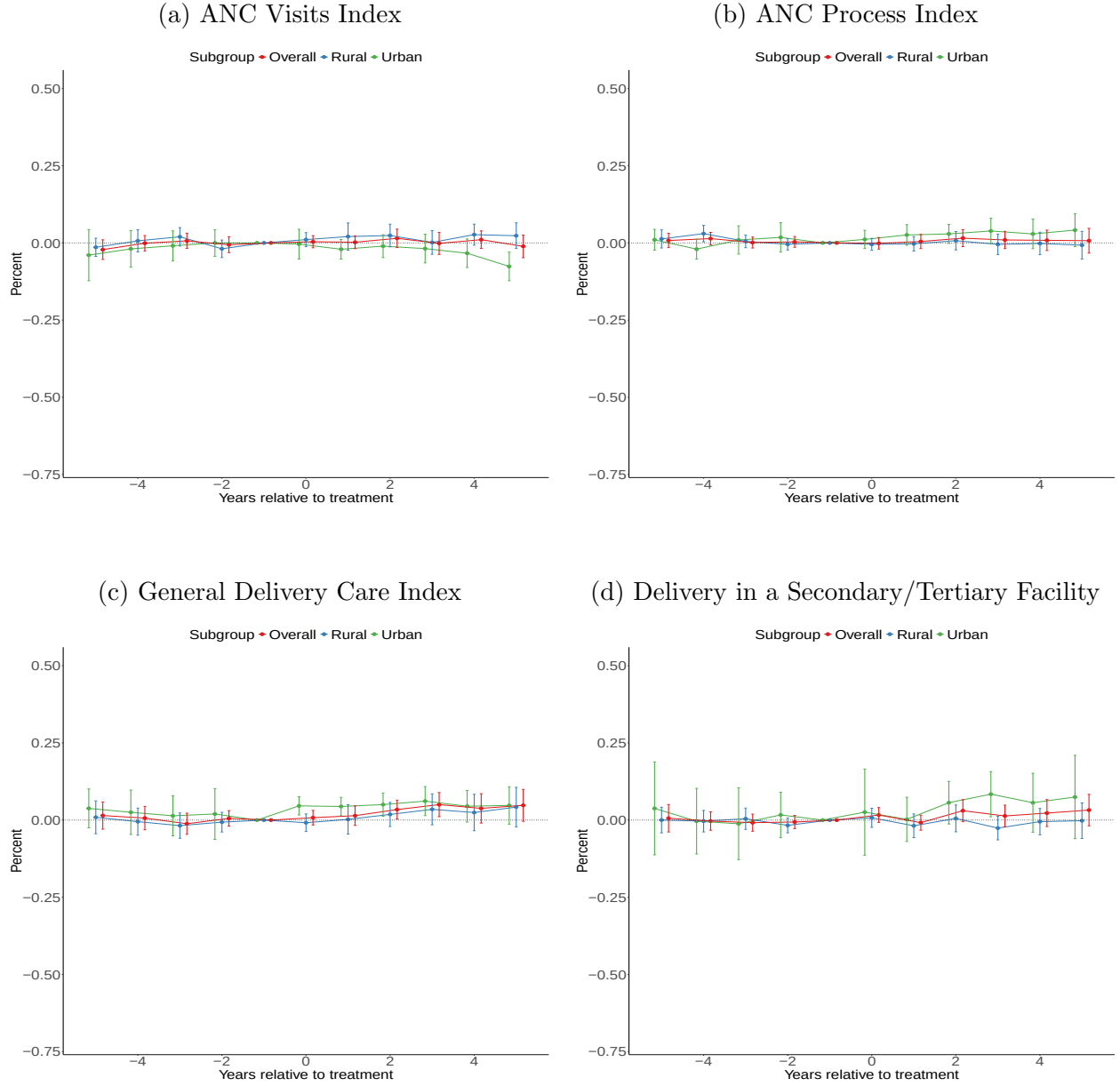
5.1.3 Healthcare Utilization

CBHI exposure increases the delivery care index by 2.8 percentage points (95% CI: 0.04, 5.5) in the overall sample, with larger gains in urban areas (4.8 percentage points, 95% CI: 2.0, 7.6). Estimated effects on ANC visits and ANC process quality are small and not statistically significant. Event study results (Figure 5) show flat pre-treatment trends, consistent with the parallel trends assumption.

These findings echo prior studies. In Thiès, [Jütting \(2004\)](#) and [Smith and Sulzbach \(2008\)](#) reported that CBHI increased access to institutional deliveries, with weaker impacts on prenatal care. Similarly, [Chankova et al. \(2008\)](#) found limited effects of MHO membership on ANC visits in Ghana, Mali, and Senegal. More recent evaluations of subsidized CBHI also found modest impacts beyond delivery care ([Bousmah et al., 2022](#); [Ly et al., 2022a](#)).

Results suggest that CBHI may reduce barriers to institutional delivery, but has more limited effects on other aspects of maternal healthcare utilization.

Figure 5. Event Study Estimates for the Healthcare Utilization Outcomes



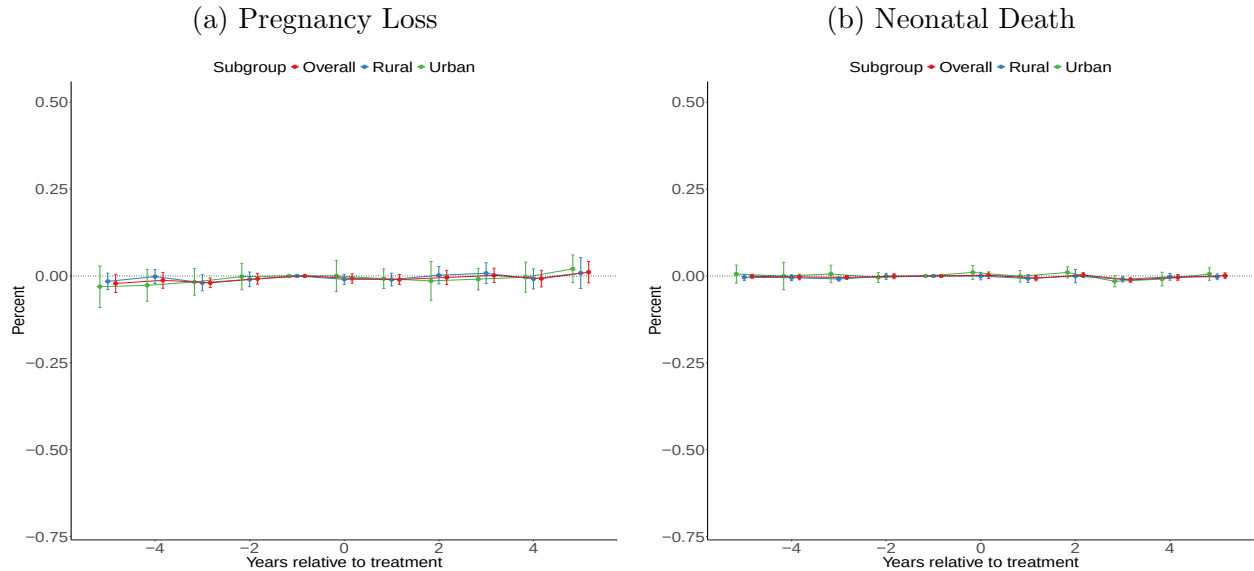
Note: This figure presents event-study estimates using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. ANC and delivery outcomes are derived from DHS birth histories and are estimated at the birth level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. To facilitate comparison across outcomes, the y-axis is standardized to range from -0.75 to 0.5 in each panel.

5.1.4 Health Outcomes

Despite gains in financial protection and modest improvements in utilization, CBHI does not significantly affect pregnancy loss or neonatal mortality. Estimated effects are close to zero in both urban and rural subsamples. Event study plots (Figures 6a and 6b) show flat trends around rollout.

These null results should be interpreted in the context of the causal chain linking insurance to health outcomes. While CBHI increased coverage and reduced financial barriers, and to a lesser extent improved delivery care, these gains did not translate into measurable improvements in pregnancy loss or neonatal mortality. This pattern is consistent with evidence that expanded access alone is insufficient to improve outcomes without adequate quality of care (Kruk et al., 2018). It also echoes findings from Das and Do (2023), who show that insurance can trigger provider behaviors, such as higher charges, unnecessary care, and uncertainty about benefit delivery, that undermine the intended effects of coverage. National audits in Senegal similarly document billing fraud among providers (Cour des Comptes, 2019), suggesting that weaknesses in implementation and quality may have limited CBHI's ability to improve health outcomes.

Figure 6. Event Study Estimates for Health Outcomes



Note: This figure presents event-study estimates using the doubly robust estimator by Callaway and Sant'Anna (2021), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. Neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes are binary. To facilitate comparison across outcomes, the y-axis is standardized to range from -0.75 to 0.5 in each panel.

5.2 Post-2013 CBHI

I also estimate effects restricting to departments first exposed to CBHI after 2013, when the program was integrated into Senegal’s UHC strategy and supported by greater government subsidies and technical assistance. A majority of departments (34 of 45, or 76%) first gained exposure in this period, providing a relevant and well-powered subsample.

Results, shown in Appendix Figure C1, closely mirror those for the full sample. Effects on insurance coverage and OOP costs remain robust, while delivery care gains are smaller and no longer statistically significant.

These findings indicate that CBHI impacts have been broadly consistent across implementation periods. Importantly, they also suggest that increased government involvement in the post-2013 era expanded program reach but did not translate into stronger effectiveness.

6 Robustness Checks

6.1 Honest Difference-in-Differences Estimates

To assess the sensitivity of my event study estimates to potential violations of the parallel trends assumption, I implement the honest difference-in-differences method proposed by [Rambachan and Roth \(2023\)](#). This approach quantifies how post-treatment estimates could be affected by deviations from trend continuity, under the assumption that such deviations are bounded by the largest fluctuation observed between adjacent pre-treatment periods.

Figure D1 presents the results. The sensitivity bands confirm that my main findings, including improvements in insurance coverage, modest gains in delivery care, and reductions in financial risk, are robust even under relaxed assumptions about trend stability. Estimated effects remain statistically distinguishable from zero across a range of plausible deviations, bolstering confidence in the main results.

6.2 Placebo Estimates

To address potential threats to identification, including compositional changes and time-varying confounders, I estimate placebo regressions using outcomes that are not plausibly affected by CBHI. These outcomes, drawn from the DHS women’s module, include women’s and partners’ education levels, employment status, receipt of a fieldworker visit, and self-reported distance to the nearest health facility. Significant changes in these variables could indicate compositional shifts in the sample or broader shocks coinciding with CBHI rollout.

As shown in Figure D2, placebo outcomes generally do not change following CBHI introduction. While I observe some differences for age and fieldworker visits, the corresponding

confidence intervals largely include zero. Overall, these results support the validity of the parallel trends assumption and mitigate concerns that estimated effects are driven by unobserved shocks or compositional changes.

6.3 Sensitivity Analyses

I conduct three additional sensitivity analyses (SAs), each designed to address a distinct concern about exposure classification or model specification. Appendix Figures [D3–D5](#) present the results.

SA 1: Excluding departments with incomplete rollout data. In four departments (Tivaouane, Mbour, Thiès, and Fatick), administrative records do not fully document the year MHOs began operating. Missing rollout dates could lead to exposure misclassification. I re-estimate my models excluding these departments. Estimates remain consistent with the baseline, indicating that incomplete rollout data do not meaningfully bias results.

SA 2: Restricting to post-2009 data. DHS cluster displacement protocols changed in 2009, constraining random displacement to remain within department boundaries. In earlier surveys, displacement may have crossed departmental boundaries, creating potential misclassification. Restricting to post-2009 DHS rounds, when displacement was constrained and boundaries were stable, yields estimates very similar to the full sample. This suggests GPS misclassification is unlikely to drive results.

SA 3: Restricting to departments with synchronized rollout. Because MHOs were often introduced at different times within a department, departmental exposure measures may aggregate substantial within-department variation. To address this, I restrict to the subset of 10 departments where all MHOs launched in the same year. Estimates differ modestly from the baseline (e.g., increased ANC visits in rural areas, declines in ANC process quality in urban areas), but are imprecise given the smaller sample. Confidence intervals overlap with baseline estimates, suggesting that the differences are not statistically significant.

6.4 Continuous Treatment Analysis

I next explore whether my binary treatment definition masks variation in CBHI intensity. Specifically, I construct a measure of the proportion of each department’s population covered by CBHI in a given year, reflecting the cumulative expansion of MHOs weighted by population coverage. As in the binary case, I verify robustness using both pre-rollout and fixed reference year population weights.

I estimate the following TWFE model:

$$Y_{idts} = \beta_0 + \beta_1 \cdot \text{CBHI_Coverage}_{dt} + X_i + \delta_d + \phi_t + \gamma_s + \epsilon_{idts}, \quad (3)$$

where $\text{CBHI_Coverage}_{dt}$ denotes the share of the population in department d with CBHI access in year t . The coefficient β_1 captures the marginal association between CBHI coverage intensity and the outcome of interest. Unlike the binary specification, which estimates the average effect of any exposure, this specification allows for dose-response analysis of program scale-up.

Several caveats limit causal interpretation. Time-varying confounders (such as concurrent investments in health infrastructure) may correlate with both CBHI expansion and outcomes. The CBHI coverage measure also relies on administrative rollout dates and projections, introducing potential measurement error, which would likely attenuate effects. Finally, the linearity assumption may overlook threshold effects, and TWFE models are subject to weighting issues even with non-negative weights (Callaway and Sant’Anna, 2021; Callaway et al., 2024). I therefore interpret these results descriptively.

Appendix Figure D6 shows limited effects on insurance coverage, with wide confidence intervals, and moderate but imprecise improvements in financial protection and utilization. Gains appear somewhat larger in rural areas, in contrast to the binary specification. Confidence intervals overlap across models, suggesting broadly consistent patterns in direction and magnitude.

To extend this analysis, I implement the estimator proposed by de Chaisemartin and D’Haultfoeuille (2024), which generalizes DiD to settings with intertemporal variation in treatment intensity. Unlike standard TWFE, this approach constructs interpretable averages of treatment effects across time and units, mitigating concerns about negative weights and heterogeneity bias. I aggregate data to the department-year level as required, reducing sample size to 45 departments observed over time.

Figure D7 displays the results. Point estimates are broadly aligned in direction with the TWFE model, reinforcing the overall conclusion that CBHI improves financial protection and delivery care quality, though modestly and unevenly across settings. Wider confidence intervals reflect the smaller sample, especially for first-stage and cost outcomes, but the consistency in direction supports the robustness of my main findings.

7 Conclusion

This study provides the first nationwide, quasi-experimental evaluation of CBHI in Senegal. I assemble new administrative data on the rollout of MHOs and link them to repeated cross-sectional household and facility surveys spanning nearly three decades. By exploiting staggered implementation across departments and applying difference-in-differences methods, I estimate the causal effects of CBHI exposure on insurance coverage, financial protection, healthcare utilization, and maternal and neonatal health outcomes.

The results show that CBHI modestly increased insurance coverage and reduced exposure to high out-of-pocket costs. Effects on utilization were limited, and I find no measurable impacts on health outcomes. Interpreted along the causal chain, CBHI achieved gains in coverage and financial protection but did not translate these into consistent improvements in service use or health. Effects were concentrated in urban areas, underscoring persistent challenges with take-up, risk pooling, and implementation capacity. These findings suggest that voluntary, small-scale CBHI schemes face structural barriers such as adverse selection, weak administrative capacity, and high costs, which constrain their effectiveness as a pathway to universal health coverage.

Several limitations merit attention. First, treatment exposure is based on reported MHO launch years, which may be imprecise, especially for older schemes, and do not reflect phased rollouts or temporary suspensions. Second, exposure is defined at the aggregate level and cannot capture individual enrollment or service use, potentially introducing measurement error. Third, repeated cross-sectional data prevent me from following individuals over time to study dynamics. Fourth, the analysis focuses on maternal and neonatal health outcomes; effects on other conditions or groups remain unexamined. Finally, data limitations preclude more granular heterogeneity analyses to identify for whom CBHI is most effective. Future research should explore these dimensions and compare community-based schemes with larger-scale models.

Despite these limitations, the evidence points to the need for institutional reforms beyond voluntary CBHI. Senegal’s recent move toward departmental health insurance units (UDAMs) represents a promising shift, offering stronger administrative oversight, wider risk pools, and greater contracting leverage with providers. Evaluating the performance of these larger-scale schemes relative to traditional CBHI is a critical next step.

Overall, the findings highlight the limits of voluntary CBHI as a pathway to universal health coverage in low-resource settings. Recent evidence suggests that mandatory or bundled schemes, especially when integrated with social protection programs, may achieve greater financial protection and care-seeking gains, though parallel investments in service

quality remain essential ([Boutin et al., 2025](#)). Expanding access and reducing financial risk will therefore require not only subsidized insurance, but also stronger delivery systems, robust institutional capacity, and policies that link demand-side financing with improvements in care quality.

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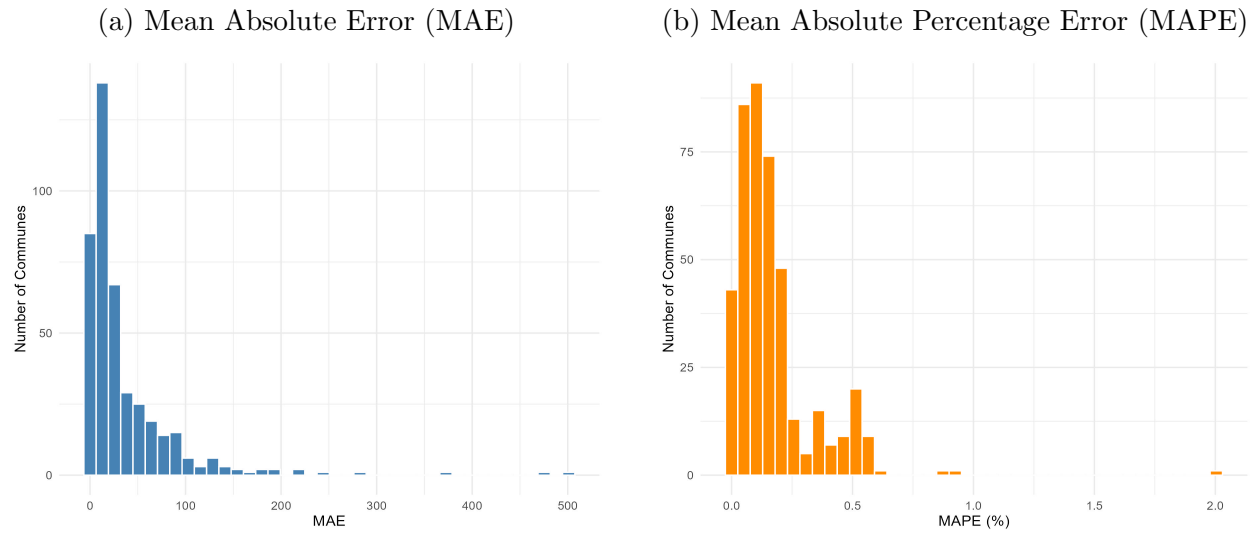
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A Appendix A: Performance of Spline-Based Population Extrapolations

Appendix Figure A1. Performance of Population Extrapolations



Note: This figure summarizes the performance of spline-based population extrapolations using data from the 424 communes that existed in 2002. Panel (a) shows the distribution of mean absolute errors (MAE) between predicted and observed population values for years with overlap. Panel (b) shows the corresponding distribution of mean absolute percentage errors (MAPE). The average mean absolute error (MAE) was 36.2, and the average mean absolute percentage error (MAPE) was 0.17%, indicating high predictive accuracy relative to commune population size.

B Appendix B: Additional Summary Statistics

Appendix Table B1. Summary Statistics of the SPA Sample, by Insurance Status

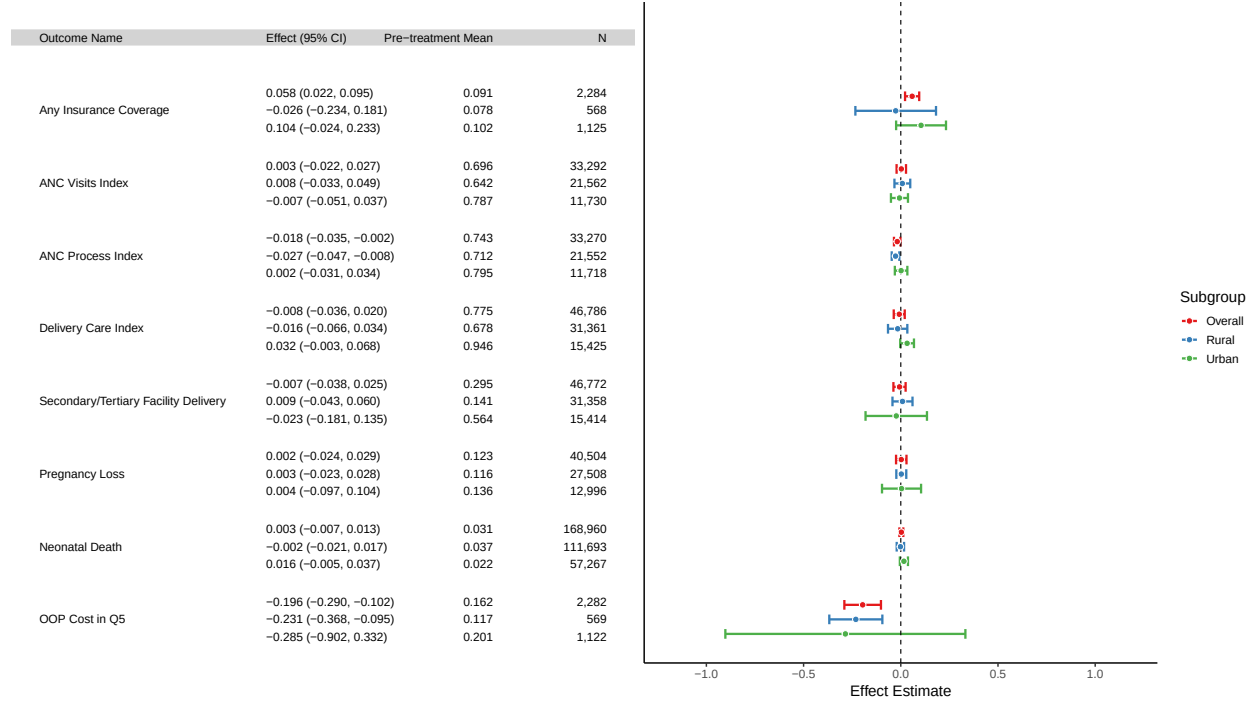
Characteristic	Overall N = 5,737	Uninsured N = 5,321	Insured N = 416
Age	27.83 (7.15)	27.67 (7.14)	29.82 (6.88)
Education level: no education	54%	56%	31%
Education level: primary	27%	26%	30%
Education level: secondary	15%	14%	28%
Education level: higher than secondary	3.9%	3.4%	11%
Location type: urban	57%	56%	78%
Facility type: health post	67%	69%	45%
Facility type: health center	25%	24%	35%
Facility type: hospital	8.2%	7.3%	20%
Facility ownership: public	93%	94%	86%
Facility ownership: non-profit	3.7%	3.6%	4.8%
Facility ownership: private	2.2%	1.7%	8.9%
Facility ownership: faith-based	0.8%	0.8%	0.2%
Service type: antenatal care	45%	46%	42%
Service type: family planning	55%	54%	58%

Abbreviations: CBHI: community-based health insurance; NGO: non-governmental organization; ANC: Antenatal care.

Note: This table summarizes characteristics of women interviewed in the SPA exit survey, both overall and stratified by insurance coverage status. Means and standard deviations (SD) are reported for continuous variables; frequencies and percentages are reported for categorical variables.

C Appendix C: Additional Subgroup Analyses

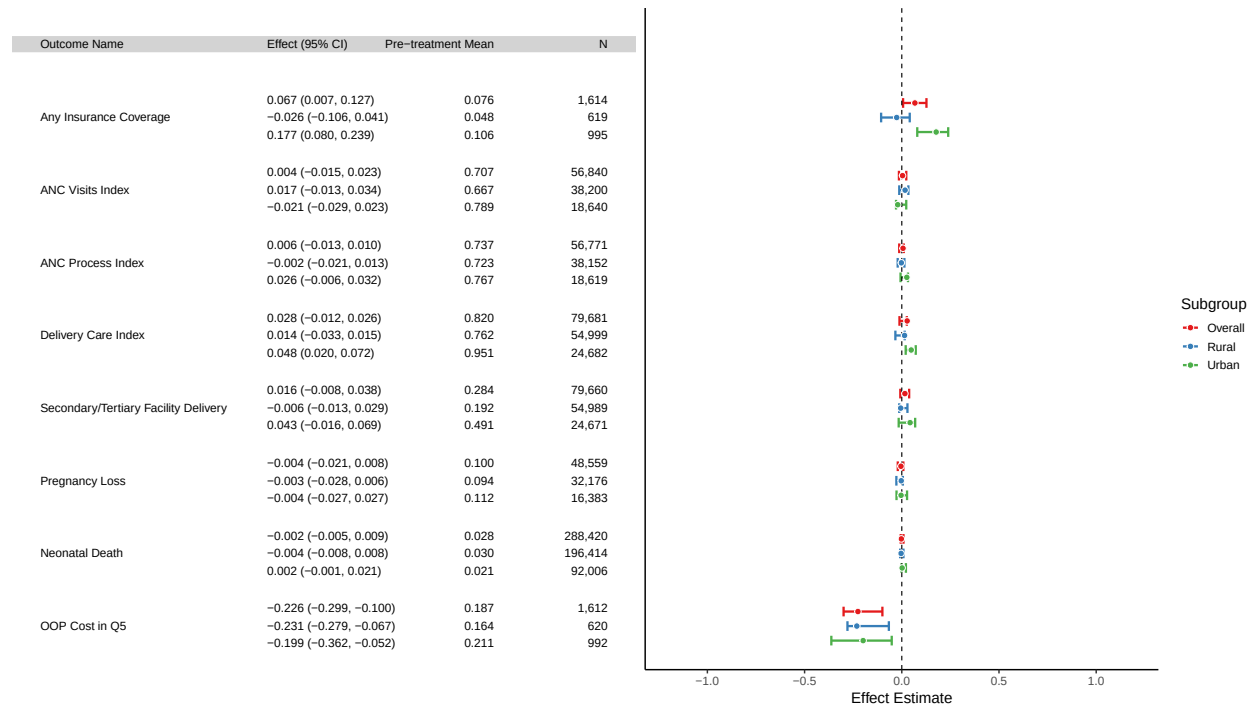
Appendix Figure C1. ATE Estimates Using Data from Departments with Exposure to CBHI After 2013



Note: This figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. This analysis is limited to departments that received CBHI exposure after 2013, when the national UHC policy was introduced. It helps isolate the impacts of the more centrally managed, government-led phase of implementation.

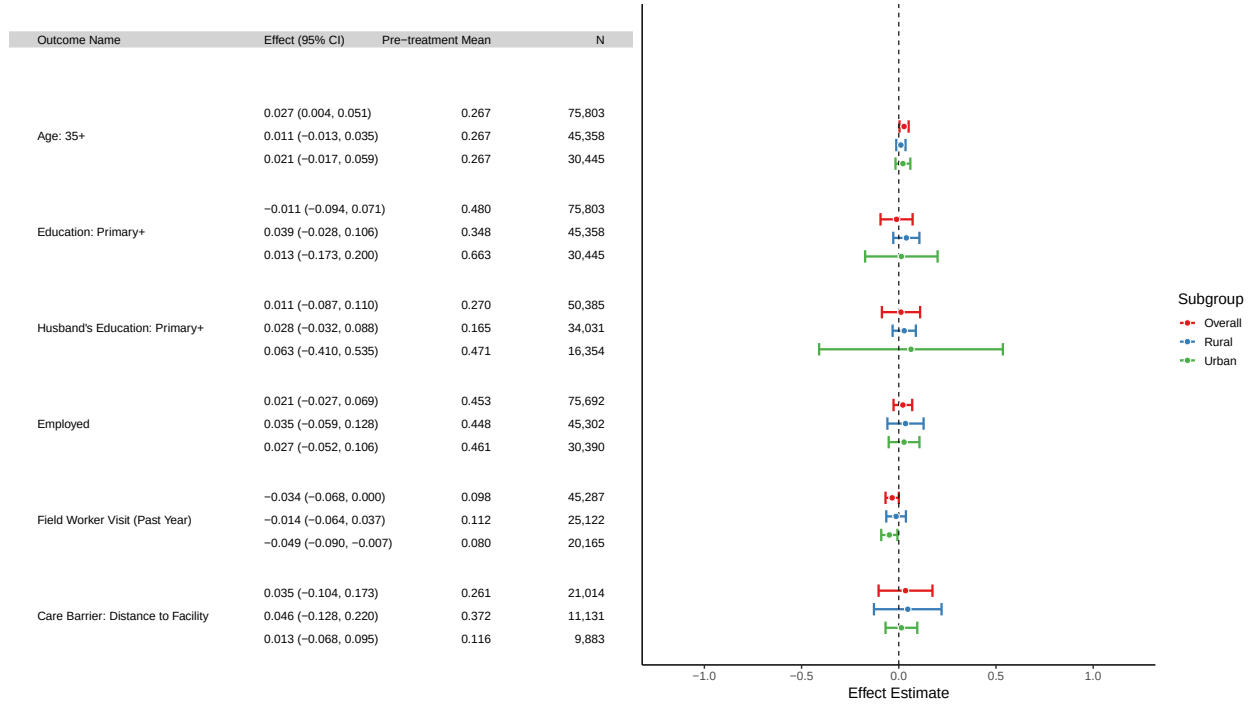
D Appendix D: Robustness Checks

Appendix Figure D1. Honest ATE Estimates



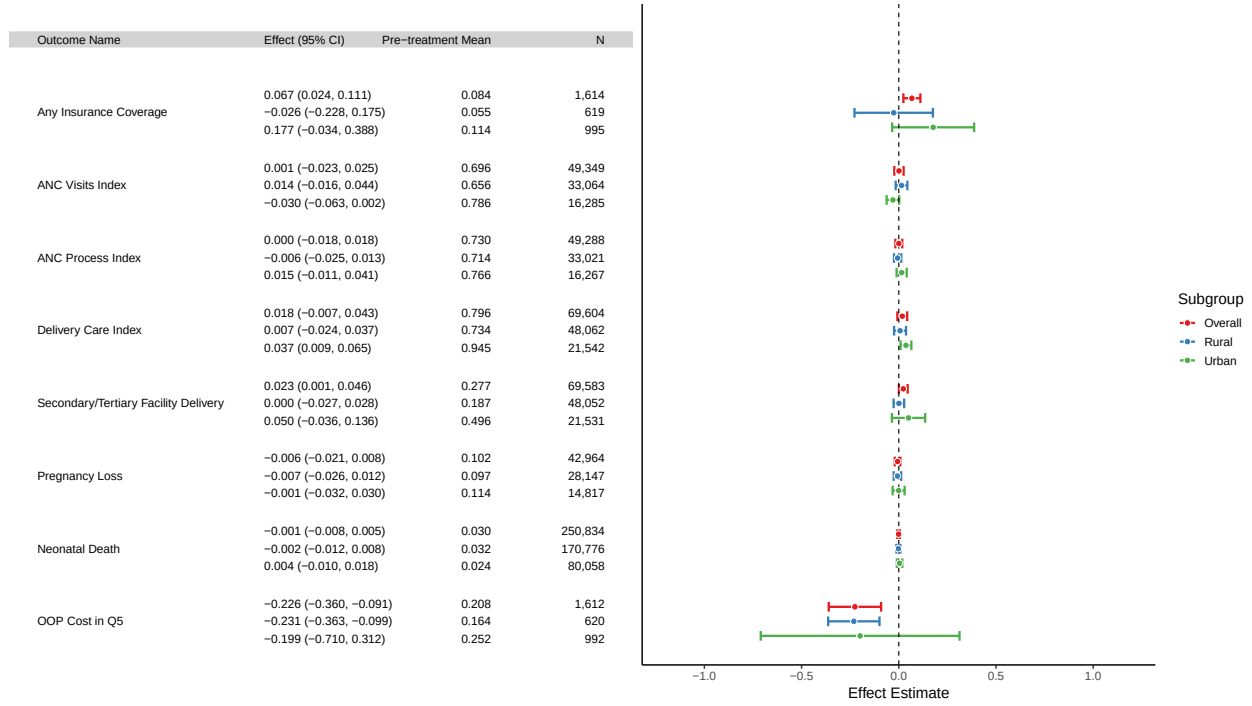
Note: This figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Confidence intervals are constructed following the approach by [Rambachan and Roth \(2023\)](#), which quantifies sensitivity to violations of the parallel trends assumption. Specifically, I assume that any post-treatment violations are no larger than the largest pre-treatment deviation observed between adjacent time periods. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and are estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year.

Appendix Figure D2. Placebo ATE Estimates



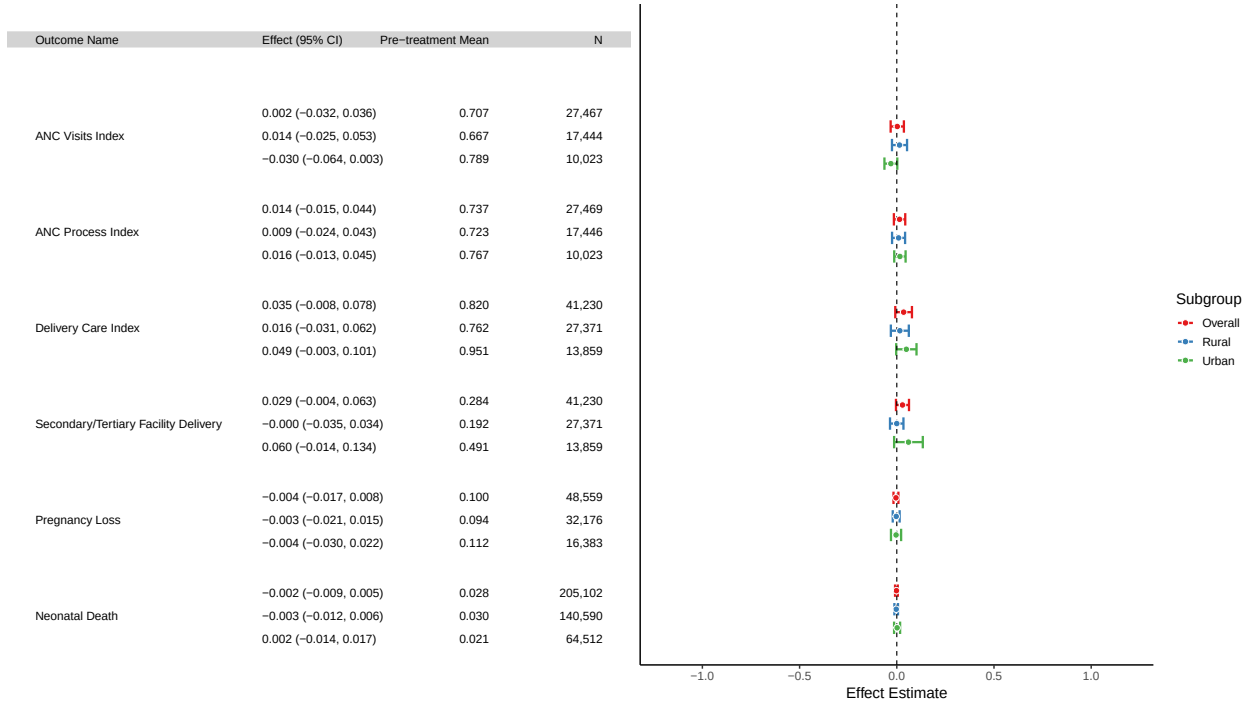
Note: This figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. The placebo outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes are binary. These outcomes—such as education, employment, and perceived distance to facilities—are not expected to be affected by CBHI exposure and help assess the validity of the identification strategy.

Appendix Figure D3. ATE Estimates for SA 1



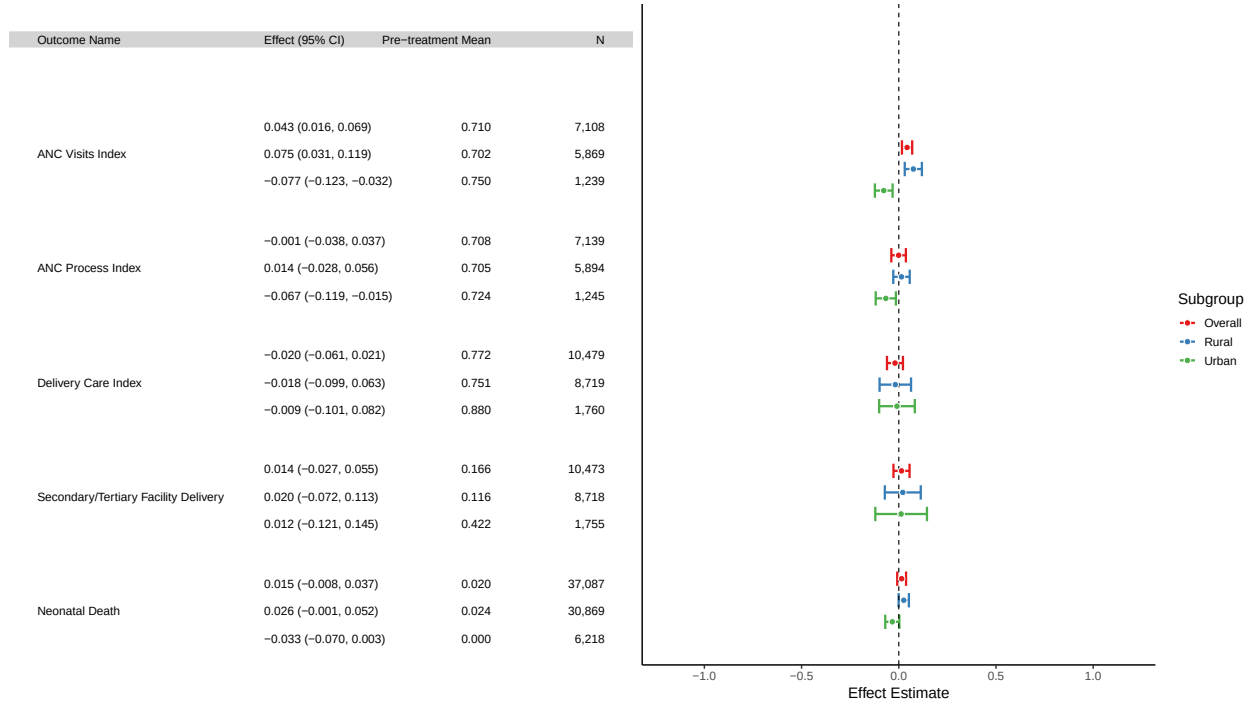
Note: This sensitivity analysis excludes four departments (Tivaouane, Mbour, Thiès, and Fatick) with incomplete information on MHO launch dates, to reduce potential exposure misclassification. The figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year.

Appendix Figure D4. ATE Estimates for SA 2



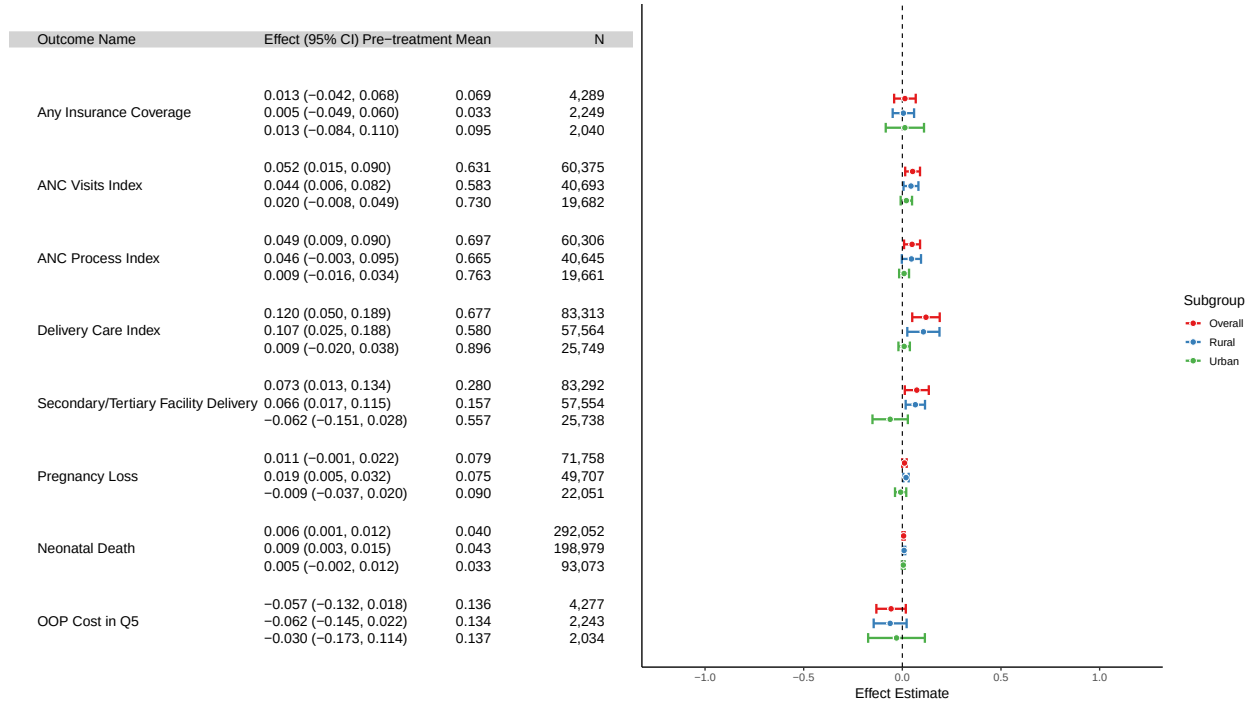
Note: This sensitivity analysis restricts the sample to post-2009 observations, when DHS cluster displacement was constrained within departments and no major redistricting occurred. This helps reduce exposure misclassification due to GPS displacement. The figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary.

Appendix Figure D5. ATE Estimates for SA 3



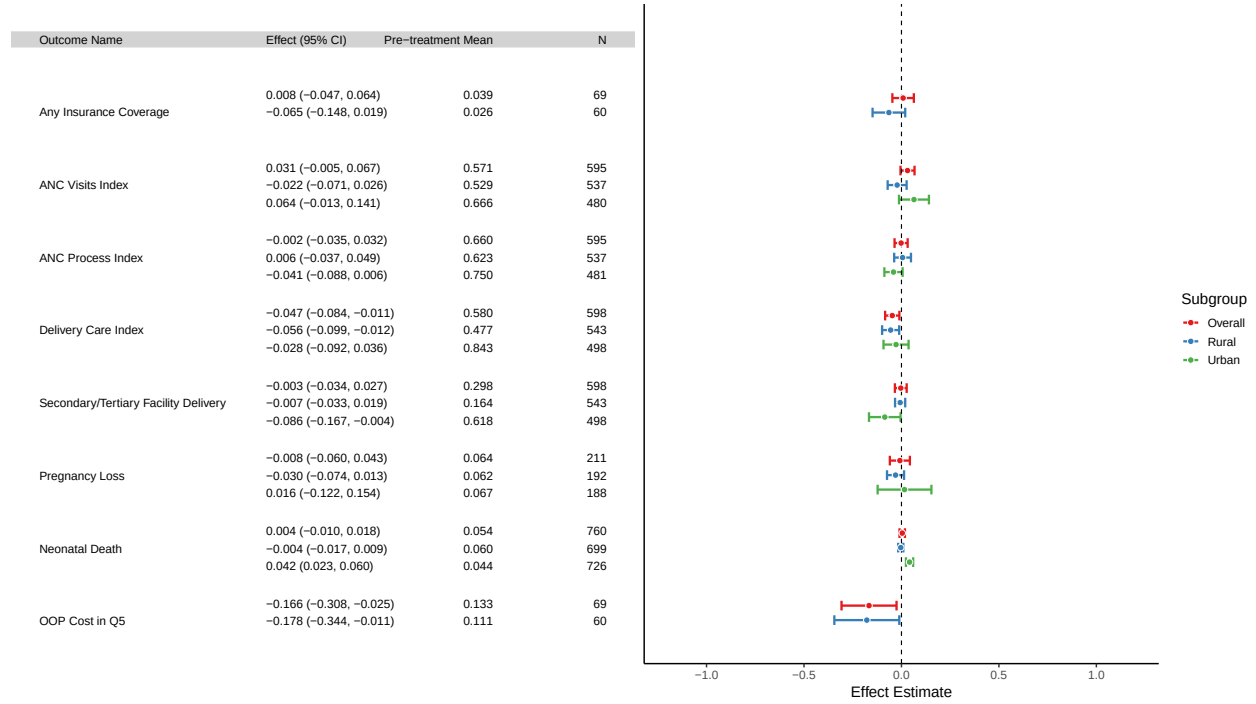
Note: This sensitivity analysis restricts the sample to departments in which all MHOs began operations in the same year, to reduce potential misclassification from within-department variation in rollout timing. The figure presents average treatment effect estimates from a difference-in-differences analysis using the doubly robust estimator by [Callaway and Sant'Anna \(2021\)](#), without covariate adjustment. Not-yet-treated observations serve as controls. For urban and rural subgroup analyses, control units are restricted to not-yet-treated observations within the same residence type. Standard errors are clustered at the department level. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in [Section 3.3](#). All other outcomes are binary.

Appendix Figure D6. Continuous ATE Estimates



Note: This figure presents average treatment effect estimates from a linear two-way fixed effects model using a continuous measure of CBHI exposure. This exposure variable reflects the proportion of each department's population with access to CBHI in a given year. Standard errors are clustered at the department-level. See Section 6.4 for additional details on the specification. First-stage and cost outcomes are based on family planning and ANC exit interviews from the SPA and are estimated at the woman level. The cost outcome refers to services received during the facility visit. ANC, delivery, and neonatal outcomes are derived from DHS birth histories and are estimated at the birth level. Pregnancy loss outcomes are drawn from the DHS contraceptive calendar and estimated at the woman-year level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as described in Section 3.3. All other outcomes are binary. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year.

Appendix Figure D7. Continuous ATE Estimates (de Chaisemartin & D'Haultfoeuille 2024).



Note: This figure presents average treatment effect estimates using the estimator proposed by [de Chaisemartin and D'Haultfoeuille \(2024\)](#), which allows for continuous treatment intensity and corrects for issues arising in staggered difference-in-differences settings. CBHI exposure is measured as the proportion of each department's population covered by CBHI in a given year. To implement the estimator, outcomes and treatment values were aggregated to the department-year level, resulting in a pseudo-panel dataset with fewer units. Consequently, confidence intervals are wider due to smaller sample sizes, and certain outcomes (e.g., urban cost and first-stage outcomes) could not be estimated due to limited data. Standard errors are clustered at the department level. All outcomes range from 0 to 1. Index measures are calculated as the proportion of components satisfied, as defined in Section 3.3. The cost outcome equals 1 if a woman's out-of-pocket expenditure for ANC or FP services falls within the top quintile of the cost distribution for her survey year. See Section 6.4 for methodological details.